Unit 9 Electricity

Topic 9.1 Parallel circuits

Exercise 9.1A Current flow in parallel circuits

1 a Parallel circuit, because there is a branch in the circuit / because current can follow more than one path / because both lamps are connected directly across the cell (and switch).

b C

2 3 + 3 = 6 (A)

3 $\frac{4}{2}$ = 2; P = 2 (A); R = 2 (A)

Exercise 9.1B Facts about parallel circuits

1 There is more than one path for current to flow in a parallel circuit.

Current divides through different parts of a parallel circuit.

2 a A_4

b A_3

c A_2 or A_3

d A_1 or A_4

3 The reading on A_1 must be the largest of all four ammeters.

The reading on A_2 must be smaller than that on A_1.

Exercise 9.1C Understanding current in parallel circuits

1 a A_1 and A_2 are equal; A_1 and A_2 are greater than A_3

b $A_4 = A_1 + A_2 + A_3$

2 a no change

b decreases

Topic 9.2 Current and voltage in parallel circuits

Exercise 9.2A Voltage

1 a volts

b The energy that the battery can supply.

2 a 1.5 V

b 1.5 V

c voltmeter

3 3; 3 (top row)

Exercise 9.2B Current and voltage

1 a Circuit copied and an ammeter anywhere in series with the other components and with a voltmeter in parallel with the buzzer.

b The voltages across each of the lamps and across the buzzer add up to the voltage across the cell.

c i decreases

ii decreases

2 a The voltage across each lamp is 4 V. Each lamp will not be at full brightness when the voltage across it is less than 12 V.

b Circuit diagram should show battery (two cells separated by a dashed line) and three lamps, each in parallel with the battery.